

BENJAMIN AGYEKUM

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with over 7 years of experience building production-grade web applications using React, Next.js, Node.js, Express and FastAPI. Architected AI-driven tutoring infrastructure leveraging LangChain and GPT-4 to serve 5M+ users and optimized database schemas for high-growth environments. Skilled in Tailwind CSS, comprehensive testing frameworks and deploying maintainable applications across frontend and backend.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, C, C++, Go, Java
- **Frontend Development:** React, Vue, TailwindCSS, HTML/CSS, Bootstrap, Webpack, Redux, React Native, Material UI, Svelte, Solid
- **Backend & APIs:** Node.js, RESTful APIs, Django, GraphQL, FastAPI, Express.js, WebSockets, Microservices, Flask, Fastify
- **AI/ML Engineering:** LangChain, LlamaIndex, RAG Systems, OpenAI APIs, PyTorch, NLP, Computer Vision
- **DevOps & Tools:** Docker, Git, VS Code, WSL2, Bash, CI pipelines, DevOps Tools, Playwright, End-to-end Testing, Full-stack Development
- **Databases & Tools:** PostgreSQL, MongoDB, MySQL, Prisma ORM, SQL, NoSQL, Jest, Pytest, Agile, Scrum

PROFESSIONAL EXPERIENCE

StudyFetch, Inc

Aug 2025 - Present

Full-Stack Software Engineer

California

- Architected AI tutoring infrastructure with OpenAI GPT-4, RAG systems, and LangChain powering Spark.E's intelligent Q&A, chat, and annotations for 5M+ users across 1M+ student conversations
- Developed cross-platform EdTech application using Next.js, React Native, and Tailwind CSS, enabling seamless document processing (PDFs, PowerPoints, videos) into personalized study tools and flashcards
- Integrated voice interaction and PDF annotation capabilities supporting StudyFetch's core features: context-aware tutoring, real-time explanations, visual document highlights, and implemented end-to-end testing with Playwright to ensure feature reliability
- Optimized MongoDB and PostgreSQL architectures to handle StudyFetch's 40% month-over-month user growth while maintaining platform performance and data retrieval efficiency

luupli.com

Apr 2021 - Mar 2024

Senior Mobile Developer(React Native)

Remote

- Served as founding mobile engineer leading React Native development for Luupli's creator-focused social commerce platform, building iOS/Android apps from beta to public launch serving 10K+ users
- Designed mobile frontend using React Native, Redux, and TypeScript, implementing video streaming and real-time notifications that enhanced user interaction and community building within the app
- Developed e-commerce infrastructure integrating secure payment APIs for creator monetization, in-app purchasing, and brand partnerships, directly supporting Luupli's mission to economically empower underserved creators
- Established mobile engineering practices including code review standards, testing protocols, and technical documentation while mentoring junior developers and collaborating across product, design, and backend teams in fast-paced startup environment

Teamalfy.com

Apr 2017 - Nov 2020

Full-Stack Developer

Remote

- Developed full-stack solutions for Teamalfy's clients in logistics, fintech, and e-commerce using Next.js, Fastify, and PostgreSQL to serve 2,000+ active users daily, enhancing user engagement and business growth
- Improved frontend application performance to achieve 90+ Lighthouse scores and sub-second Time to Interactive using Next.js SSR optimization, lazy loading, and Tailwind CSS, boosting client SEO rankings and user retention
- Constructed secure backend infrastructure with RESTful APIs, JWT authentication, and OAuth 2.0 for payment systems, processing 500+ monthly transactions without security breaches, ensuring platform reliability and client trust
- Implemented Docker containerization, CI/CD pipelines with GitHub Actions, and end-to-end testing with Playwright, cutting deployment time by 50% and sustaining 99.9% service uptime through thorough testing and monitoring

Microverse

Jan 2022 - Jun 2022

Software Developer Intern

San Francisco

- Collaborated in a global remote team to design and build an ongoing analytics dashboard for Microverse learners, delivering real-time insights on student progress, assignments, and performance metrics using React.js, D3.js.
- Integrated Node.js with Azure cloud services to build scalable backend APIs, enabling seamless data exchange and improving application performance by 25% through optimized server-side processing for a remote development environment.

EDUCATION

Colorado State University

Aug 2023 - May 2025

Master's, Electrical Engineering

Fort Collins

CERTIFICATIONS

- Machine Learning & AI Engineer, Data Structure and Algorithm, Natural Language Processing and Artificial Intelligence